Eidgenössische Technische Hochschule Zürich
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High-Performance Computing Lab for CSE **2026**

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Report for Project 2

Due date: 23 March 2026, 18:00 (Europe/Zurich)

HPC Lab for CSE 2025 — Submission Instructions

Following these instructions is mandatory: Submissions that do not comply **will not be considered**.

- **Submission Platform**: Assignments must be submitted via SAM-UP.
- **Required Files**: Your submission must include:
  - **Report**: A PDF summarizing your results and observations, based on this LaTeX template.
    - File name: `project_number_lastname_firstname.pdf`
    - You are allowed to discuss all questions with anyone you like; however: (i) your submission must list anyone you discussed problems with and (ii) you must write up your submission independently.
  - **Source Code Archive**: A **compressed tar archive**, named `project_number_lastname_firstname.tar.gz`, containing:
    - All source code files.
    - Build files and scripts (e.g., `Makefile`, batch job scripts, plot scripts, etc). If you modified any provided files, ensure they still work as expected.
    - The full LaTeX source of your report including all needed files to build the report (figures, listings, etc).
  - **Size Limit**: The total submission (PDF + archive) must not exceed **25 MB**.
  - **File Naming Rules**: Use only `ASCII` characters (avoid spaces, special symbols, and non-English letters).
- **Grading Process**: The TAs will evaluate your project based on:
  - Your report write-up.
  - Your code implementation.
  - Benchmarking your code's performance.

1. Computing π with OpenMP [20 points]

1.1. Exercise 1

In this exercise we are given a code that computes an approximation of π using numerical integration. The code uses a simple loop to compute the sum of terms that approximate the value of π. The goal of this exercise is to parallelize the code using OpenMP and to optimize it for better performance.

In the first part of the exercise it was asked to first parallelize the given code using a naive version that protects every loop update with `critical`. Here is the code snippet of the naive version that we implemented:

```
  #pragma omp parallel for
  for (uint64_t i = 0; i < N; ++i) {
    double x = (i + 0.5) * h;
    double term = 4.0 / (1.0 + x*x);

    #pragma omp critical
    pi += term;
  }
  pi *= h;

  double time = walltime() - time_start;

  int nt = omp_get_max_threads();
  printf("# pi N num_threads time [secs]\n");
  printf("%.15f %20"PRIu64" %3d %.8f\n", pi, N, nt, time);
```

The output of the naive version is as follows:

```
# pi N num_threads time [secs]
3.141592653590757 1000000000 64 640.70646905
```

We can see that the naive version is very slow. This is because every update to the shared variable `pi` is protected by a critical section, which causes a lot of contention and serialization among the threads. Each thread has to wait for the others to finish updating `pi`, which leads to a significant slowdown in performance.

In the second part in was asked to create an optimized version that accumulates locally and enters critical only once per thread. Here is the code snippet of the optimized version that we implemented:

```
  #pragma omp parallel
  {
    double sum = 0.0;

    #pragma omp for
    for (uint64_t i = 0; i < N; ++i) {
      double x = (i + 0.5) * h;
      sum += 4.0 / (1.0 + x*x);
    }

    #pragma omp critical
    pi += sum; // once per thread
  }
  pi *= h;

  double time = walltime() - time_start;

  int nt = omp_get_max_threads();
  printf("# pi N num_threads time [secs]\n");
  printf("%.15f %20"PRIu64" %3d %.8f\n", pi, N, nt, time);
```

In this version we have a local variable `sum` that is private to each thread, and we only enter the critical section once per thread to update the shared variable `pi`. This reduces the contention on the critical section and allows for better performance. The output of the optimized version is as follows:

```
# pi N num_threads time [secs]
3.141592653589771 1000000000 64 0.19858712
```

We can see that the optimized version is significantly faster than the naive version. The optimized version allows for better utilization of the available threads and reduces the overhead caused by the critical section.

We are then asked to implement a version that uses the reduction clause. Here is the code snippet of the version that uses the reduction clause:

```
  #pragma omp parallel for reduction(+:pi)
#pragma omp parallel for reduction(+:sum)
  for (uint64_t i = 0; i < N; ++i) {
    double x = (i + 0.5)*h;
    sum += 4.0 / (1.0 + x*x);
  }
  pi = sum*h;
  double time = walltime() - time_start;

  int nt = omp_get_max_threads();
  printf("# pi N num_threads time [secs]\n");
  printf("%.15f %20"PRIu64" %3d %.8f\n", pi, N, nt, time);
```

In this version we use the `reduction` clause to automatically handle the accumulation of the variable `pi` across threads. The reduction clause takes care of creating a private copy of the variable for each thread and then combining the results at the end of the parallel region. This allows for a more concise and efficient implementation. The output of the version that uses the reduction clause is as follows:

```
# pi N num_threads time [secs]
3.141592653589771 1000000000 64 0.11437800
```

We can see that the optimized version and the version that uses the reduction clause are significantly faster than the naive version. The reduction clause version is also more concise and easier to read, making it a good choice for this type of problem.
In table 1 we summarize the results of the different versions of the code.

Version	pi	num_threads	time [secs]
Serial	3.141592653589971	1	1.88825762
Naive	3.141592653590757	64	640.706
Optimized	3.141592653589771	64	0.19858712
Reduction	3.141592653589771	64	0.11437800

Table 1: Results of the different versions of the code for computing π

1.2. Exercise 2

In the second part of the exercise we are asked to perform a weak and strong scaling study.

1.2.1. Strong scaling study

In this context a strong scaling study is performed by keeping the problem size fixed and increasing the number of threads. The goal is to see how the performance of the code improves as we increase the number of threads. In our case, we can keep the value of N fixed and measure the execution time for different numbers of threads. We can then plot the speedup and efficiency as a function of the number of threads to analyze the strong scaling behavior of our code.
Here are outputs.
Strong naive

```
  # pi N num_threads time [secs]
3.141592653589634 1000000000 1 6.93277019
3.141592653589634 1000000000 2 76.46882527
3.141592653589634 1000000000 4 197.49973289
3.141592653589634 1000000000 8 283.80093214
3.141592653589634 1000000000 16 403.18691688
3.141592653589634 1000000000 32 501.40493621
3.141592653589634 1000000000 64 678.71737772
```

Strong optimized)

```
# pi N num_threads time [secs]
3.141592653589771 1000000000 1 2.26213344
3.141592653589771 1000000000 2 1.26299218
3.141592653589771 1000000000 4 0.67451669
3.141592653589771 1000000000 8 0.33567476
3.141592653589771 1000000000 16 0.17712136
3.141592653589771 1000000000 32 0.10922673
3.141592653589771 1000000000 64 0.10740197
```

Strong reduction)

```
  # pi N num_threads time [secs]
3.141592653589772 1000000000 1 2.19981237
3.141592653589772 1000000000 2 1.26300334
3.141592653589772 1000000000 4 0.67434969
3.141592653589772 1000000000 8 0.33552001
3.141592653589772 1000000000 16 0.17785895
3.141592653589772 1000000000 32 0.12054630
3.141592653589772 1000000000 64 0.07776448
```

Weak naive)

```
  3.141592653589632 100000 1 0.00118749
3.141592653589632 200000 2 0.02662989
3.141592653589632 400000 4 0.13208618
3.141592653589632 800000 8 0.22754601
3.141592653589632 1600000 16 0.62820553
3.141592653589632 3200000 32 2.44359762
3.141592653589632 6400000 64 5.08367890
```

Weak optimized)

```
  # pi N num_threads time [secs]
3.141592653589791 100000 1 0.00026265
3.141592653589791 200000 2 0.00038406
3.141592653589791 400000 4 0.00040525
3.141592653589791 800000 8 0.00053178
3.141592653589791 1600000 16 0.00072227
3.141592653589791 3200000 32 0.01441353
3.141592653589791 6400000 64 0.02746264
```

Weak reduction)

```
  # pi N num_threads time [secs]
3.141592653589791 100000 1 0.00025960
3.141592653589791 200000 2 0.00037925
3.141592653589791 400000 4 0.00044645
3.141592653589791 800000 8 0.00058488
3.141592653589791 1600000 16 0.00804680
3.141592653589791 3200000 32 0.01407022
3.141592653589791 6400000 64 0.02701498
```

2. The Mandelbrot set using OpenMP [20 points]

2.1. Exercise 1 and 2

Here we are asked to complete the serial version of the Mandelbrot set code in the missing code part and to count the total number of iterations in order to correctly compute the benchmark statistics.
In the following code snippet we show the completed version of the serial code for computing the Mandelbrot set:

```
double time_start = walltime();
  // do the calculation
  cy = MIN_Y;
  for (j = 0; j < IMAGE_HEIGHT; j++) {
    cx = MIN_X;
    for (i = 0; i < IMAGE_WIDTH; i++) {
      x = cx;
      y = cy;
      x2 = x * x;
      y2 = y * y;
      // compute the orbit z, f(z), f^2(z), f^3(z), ...
      // count the iterations until the orbit leaves the circle |z|=2.
      // stop if the number of iterations exceeds the bound MAX_ITERS.
      int n = 0;

      while (x2 + y2 <= 4.0 \&\& n < MAX\_ITERS) {
        y = 2.0 * x * y + cy;
        x = x2 - y2 + cx;
        x2 = x * x;
        y2 = y * y;
        n++;
      }
       nTotalIterationsCount += n;
      // plot the number of iterations at point (i, j)
      int c = ((long)n * 255) / MAX_ITERS;
      ppm_plot(&pPpm, i, j, c, c, c);
      cx += fDeltaX;
  }
    cy += fDeltaY;
  }
  double time_end = walltime();
```

In the while loop, we are iterating through the orbit of the point c under the function $f(z) = z^2 + c$. The variables x and y represent the real and imaginary parts of the complex number z, respectively. The equations used in the while loop are derived from the definition of complex multiplication and addition. Specifically, the function $f(z)$ can be expressed in terms of its real and imaginary components as follows:ù

$$\begin{aligned} f(z) &= z^2 + c \\ &= (x + yi)^2 + (cx + cyi) \\ &= (x^2 - y^2 + cx) + (2xy + cy)i \end{aligned}$$

But why does this hold also for $f_z(c)^3, ...$? This is because the function $f(z)$ is defined recursively, and each iteration of the function applies the same transformation to the complex number z. Therefore, when we compute $f^2(z)$, we are applying the function f to the result of $f(z)$, which means that we are applying the same equations for the real and imaginary parts at each iteration. This recursive nature of the function allows us to use the same equations for each iteration in the while loop, regardless of how many times we apply the function. From this, we can see that the real part of f(z) is given by $x^2 - y^2 + cx$, and the imaginary part is given by $2xy + cy$. Therefore, in the while loop, we update the values of x and y according to these equations to compute the next point in the orbit. The loop continues until the magnitude of z (which is $\sqrt{x^2 + y^2}$) exceeds 2, indicating that the orbit is unbounded, or until we reach the maximum number of iterations, which indicates that the point c is likely to belong to the Mandelbrot set.
Finally, we are asked to count the total number of iterations in order to correctly compute the benchmark

statistics. This is done by introducing a variable nTotalIterationsCount that accumulates the number of iterations for each point in the complex plane. This variable is incremented by n, which represents the number of iterations for the current point (i, j). By keeping track of the total number of iterations, we can compute the benchmark statistics more accurately, as it gives us a measure of the computational effort required to compute the Mandelbrot set for the given parameters.

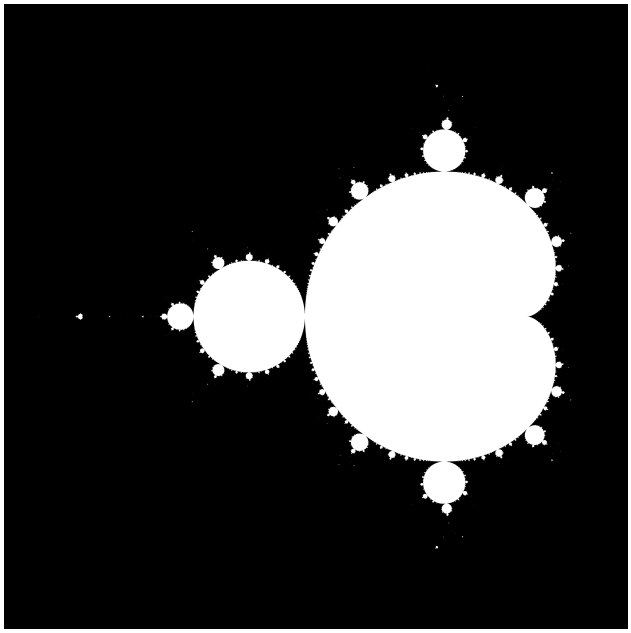


Figure 1: Serial Mandelbrot Set

2.2. Exercise 3

In the third part of the exercise we are asked to parallelize the Mandelbrot set code using `OpenMP`. The goal is to improve the performance of the code by utilizing multiple threads to compute the Mandelbrot set in parallel. Here is the code snippet of the parallelized version of the Mandelbrot set code using `OpenMP`:

```
int main(int argc, char **argv) {
ppm_data_t pPpm;
ppm_create(&pPpm, IMAGE_WIDTH, IMAGE_HEIGHT);

double fDeltaX = (MAX_X - MIN_X) / (double)IMAGE_WIDTH;
double fDeltaY = (MAX_Y - MIN_Y) / (double)IMAGE_HEIGHT;

long nTotalIterationsCount = 0;

double time_start = walltime();
// do the calculation
#pragma omp parallel for reduction(+:nTotalIterationsCount) schedule(dynamic)
for (long j = 0; j < IMAGE_HEIGHT; j++) {
  double cy = MIN_Y + j * fDeltaY;

  for (long i = 0; i < IMAGE_WIDTH; i++) {
    double cx = MIN_X + i * fDeltaX;

    double x = cx;
    double y = cy;
    double x2 = x * x;
    double y2 = y * y;

    int n = 0;
    while (x2 + y2 <= 4.0 && n < MAX_ITERS) {
```

```
      y = 2.0 * x * y + cy;
      x = x2 - y2 + cx;
      x2 = x * x;
      y2 = y * y;
      n++;
    }

    nTotalIterationsCount += n;

    int c = ((long)n * 255) / MAX_ITERS;
    ppm_plot(&pPpm, i, j, c, c, c);
  }
}
double time_end = walltime();
```

Metric	Serial Version	Parallel Version
Total time [seconds]	330.726	6.70717
Total number of iterations	113610974266	113638786651
Average time per pixel [seconds]	1.97128e-05	3.99779e-07
Average time per iteration [seconds]	2.91104e-09	5.90219e-11
Iterations per second	3.4352e+08	1.69429e+10
MFlop/s	2748.16	135543

Table 2: Comparison of the serial and parallel versions of the Mandelbrot set code

The results (see table 2) show a significant performance improvement achieved by parallelizing the code using `OpenMP`. The total time taken to compute the Mandelbrot set is reduced from 330.726 seconds in the serial version to 6.70717 seconds in the parallel version, which is a substantial decrease in execution time. The total number of iterations is slightly higher in the parallel version, which could be due to differences in the order of execution and how the iterations are distributed among the threads. The average time per pixel and average time per iteration are also significantly reduced in the parallel version, indicating that the computation is much faster when utilizing multiple threads. The iterations per second and MFlop/s metrics show a dramatic increase in performance, with the parallel version achieving 1.69429e+10 iterations per second and 135543 MFlop/s compared to 3.4352e+08 iterations per second and 2748.16 MFlop/s in the serial version. This performance improvement can be attributed to the ability of `OpenMP` to effectively utilize the available CPU cores, allowing for concurrent execution of the computations required to determine whether each point belongs to the Mandelbrot set. By distributing the workload across multiple threads, the parallel version can significantly reduce the time required to compute the Mandelbrot set, especially for larger image sizes and higher iteration counts.

2.2.1. Strong scaling analysis

To analyze the strong scaling of the parallelized Mandelbrot set code, we can measure the execution time for different numbers of threads while keeping the problem size fixed. This allows us to see how well the code scales with increasing thread count.
We also calculate the speedup as the ratio of the execution time of the serial version to the execution time of the parallel version for each thread count. In figure 2 we show the plots of the strong scalying analysis. As we can see the execution times halves every time we double the number of threads, which indicates that the code scales well with increasing thread count. In figure 3 instead we plot the theoretical speedup with the one computed from the simulations. The ideal speedup is computed via 1

$$S_{ideal}(n) = n \tag{1}$$

where n is the number of threads. The actual speedup is computed as shown in 2

$$S_{actual}(n) = \frac{T_{serial}}{T_{parallel}(n)} \tag{2}$$

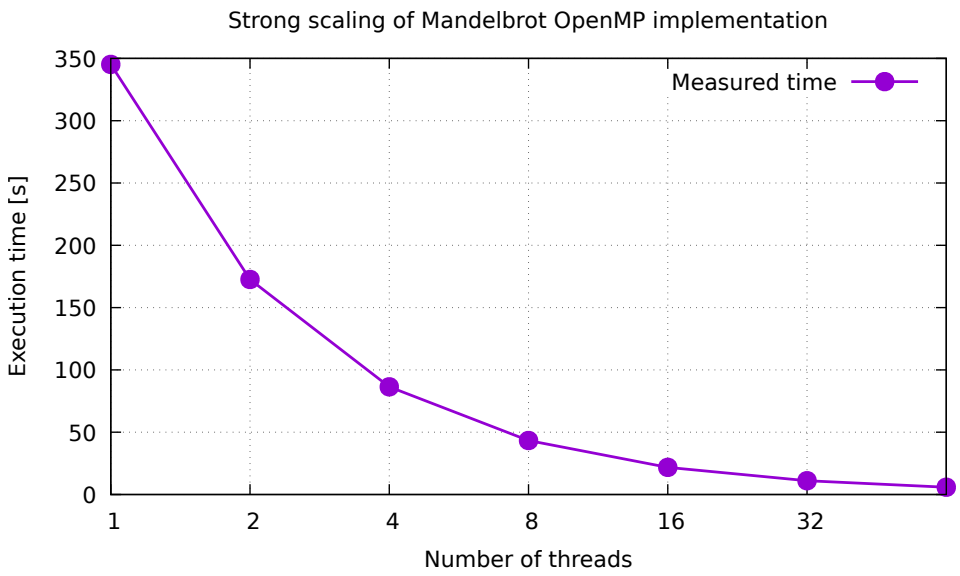


Figure 2: Strong scaling analysis of the parallelized Mandelbrot set code

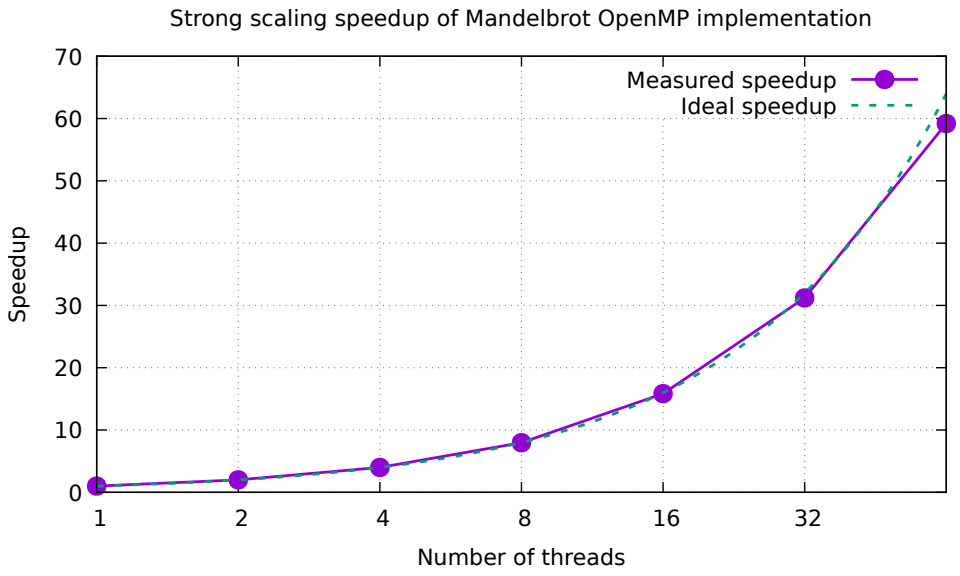


Figure 3: Speedup of the parallelized Mandelbrot set code compared to the ideal speedup

3. Bug hunt [10 points]

3.1. File 1: omp_bug1.c

The program attempts to parallelize the loop that computes

$$c[i] = a[i] + b[i]$$

using the OpenMP `parallel for` construct. However, the code contains a syntax error that prevents compilation and also reflects a conceptual misuse of the directive.

The first problem is related to the use of the OpenMP directive `#pragma omp parallel for`. This combined directive creates a parallel region and distributes the iterations of the following loop among the threads. For this reason, the directive must be followed *immediately* by a `for` loop. In the program, however, the directive is followed by a compound block:

```
#pragma omp parallel for shared(a, b, c, chunk) private(i, tid) \
          schedule(static, chunk)
{
  tid = omp_get_thread_num();
  for (i = 0; i < N; i++) {
    ...
  }
}
```

This structure is invalid because the `parallel for` directive must be attached directly to the loop it parallelizes.

The second issue concerns the variable `tid`. The programmer wants to print which thread executes each iteration, but the statement

```
tid = omp_get_thread_num();
```

is placed outside the loop that the directive is supposed to parallelize. Because the directive requires the loop to appear immediately after it, this statement breaks the required structure of the construct.

To fix the problem, the loop must appear directly after the OpenMP directive. The thread identifier should instead be retrieved inside the loop body (or inside a parallel region if the `parallel` and `for` constructs are separated). In this way, each thread can correctly obtain its ID and print the information for the iterations it executes.

3.2. File 2: omp_bug2.c

The program contains errors related to the incorrect handling of `shared` and `private` variables inside the OpenMP parallel region.

The variables `tid`, `i`, and `total` are declared outside the `#pragma omp parallel` region and therefore are *shared* by default among all threads. This leads to race conditions and incorrect program behaviour.

First, the variable `tid` is shared, although each thread assigns its own thread ID through

```
tid = omp_get_thread_num();
```

Since all threads write to the same memory location, the value of `tid` can be overwritten by other threads. As a result, the value stored in `tid` is not guaranteed to correspond to the correct thread.

A more serious problem concerns the variable `total`. Inside the parallel region the code initializes

```
total = 0.0;
```

and then performs a parallel loop

```
#pragma omp for schedule(dynamic, 10)
for (i = 0; i < 1000000; i++)
    total = total + i * 1.0;
```

Because `total` is shared, all threads update the same variable concurrently. This produces a race condition and the resulting sum is incorrect and non-deterministic. Moreover, each thread resets the shared variable to zero before the loop, which introduces additional data races.

Finally, the statement

```
printf("Thread %d is done! Total= %e\n", tid, total);
```

is located outside the parallel region. Consequently, it is executed only once by the main thread, even though the programmer likely intended each thread to report completion.

The problem can be fixed by ensuring correct data scoping. The variable `tid` should be declared *private* so that each thread keeps its own thread identifier. The summation should be implemented using an OpenMP `reduction` on `total`, which safely combines the partial sums computed by different threads. If each thread should report completion, the corresponding `printf` statement should be placed inside the parallel region.

3.3. File 3: `omp_bug3.c`

The main problem in this program is the incorrect use of the `barrier` directive inside the function `print_results`. In OpenMP, a barrier must be encountered by *all* threads of the team, or by none of them. Here, however, `print_results` is called only by the threads that actually execute one of the two `sections`. Since there are only two sections, at most two threads will enter the function, while the other threads in the parallel region will skip it entirely. As a consequence, only a subset of threads reaches the barrier placed at the end of `print_results`, which leads to an illegal synchronization pattern and can cause the program to hang at run time.

A second issue is that the intent of having "clean output" is only partially achieved. The `critical` region correctly prevents the printed array contents from being interleaved, so the output of one thread is kept together. However, the statement

```
printf("Thread %d done and synchronized.\n", tid);
```

is placed *after* the barrier and outside the critical region, so even if the barrier were legal, these messages could still appear in overlapping or non-deterministic order with respect to other threads' output.

There is also a conceptual inconsistency in combining `sections nowait` with the later synchronization logic. The `nowait` clause removes the implicit barrier at the end of the `sections` construct, but then the code tries to enforce synchronization manually inside `print_results` using a barrier, which is not valid for the reason explained above. If synchronization after the sections is desired, it should be done at a point that is reached by every thread in the parallel region.

To fix the code, the barrier inside `print_results` should be removed. Any barrier should instead be placed in the main parallel region, at a location where all threads are guaranteed to encounter it. Moreover, if the programmer wants all output messages to appear cleanly and without overlap, every printing operation that must remain grouped should be enclosed in the same `critical` region, or alternatively all printing should be reorganized so that only one thread prints at a time in a controlled way. Finally, if synchronization at the end of the `sections` construct is required, the simplest correction is to remove the `nowait` clause and rely on the implicit barrier provided by `sections`.

In summary, the critical bug is the illegal barrier inside `print_results`, because not all threads execute that function. This may cause a deadlock. The proper fix is to move synchronization points outside the function and ensure that every barrier is reached by all threads in the team.

3.4. File 4: `omp_bug4.c`

The problem is caused by *stack overflow* in the OpenMP region. The key issue is that the array

```
real(8) ::  e_in(nmax)
```

is declared inside the main program and the parallel region uses

```
!$omp parallel default(firstprivate).
```

Because of `firstprivate`, each OpenMP thread receives its own private copy of `e_in`. Since

$$n_{\max} = 202000,$$

this array alone occupies about

$$202000 \times 8 \approx 1.6 \text{ MB}$$

per thread. These private copies are typically placed on the thread stack. If the available stack size is too small, entering the parallel region causes a segmentation fault.

There is a second large object involved:

```
double complex ::  eTDSE(ntmax)
```

inside the module, declared as

```
!$omp threadprivate(eTDSE).
```

This means that each thread also has its own copy of `eTDSE`. A `double complex` element usually requires 16 bytes, so this array takes about

$$202000 \times 16 \approx 3.2 \text{ MB}$$

per thread. Therefore, with OpenMP enabled, the memory required per thread becomes quite large. The crash appears only when both arrays are effectively used in the parallel region, which is why removing one of the `print` statements may make the program run.

The code is not wrong from a syntax point of view, but it is wrong from a memory-management perspective. The main mistake is using `default(firstprivate)` in a region where a large local array such as `e_in` exists. This forces unnecessary duplication of a large array on each thread stack. In addition, `threadprivate(eTDSE)` also creates one copy per thread, which further increases memory pressure.
So the real bug is the *combination of large thread-local data and insufficient thread stack size.*

A first possible fix is to avoid making `e_in` firstprivate if this is not needed. Since the array is only read in the loop, it should be shared instead of private. In practice, one should replace the overly broad

```
default(firstprivate)
```

with a more careful data-sharing specification, for example making `e_in` shared.

A second fix is to increase the stack size. On Unix-like systems this can be done for the main thread with

```
ulimit -s
```

and for OpenMP worker threads with environment variables such as `OMP_STACKSIZE`. With Intel OpenMP, `KMP_STACKSIZE` can also be used. This does not remove the underlying inefficiency, but it may prevent the crash.

Another possible improvement is to avoid storing such large arrays on the stack. Allocatable arrays or heap-based storage are often preferable for large data structures in OpenMP programs. This reduces the risk of stack overflow and makes memory usage more controllable.

The segmentation fault is caused by insufficient stack memory, not by an invalid OpenMP construct. The array `e_in` becomes private to each thread because of `default(firstprivate)`, and `eTDSE` is also duplicated because it is `threadprivate`. The resulting per-thread memory usage is too large for the available stack. The proper fix is to avoid unnecessary privatization of large arrays, and, if needed, increase the OpenMP thread stack size or move the data to heap allocation.

4. Parallel histogram calculation using `OpenMP` [15 points]

5. Parallel loop dependencies with `OpenMP` [15 points]

6. Quicksort using `OpenMP` tasks [20 points]